

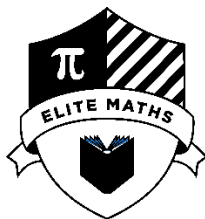


Year-9-nsw-5.15.25.3-sample-questions

Maths (High School - Australia)



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2021 YEAR 9 (5.1) MATHEMATICS

Written examination

Reading time: 15 minutes

Writing time: 3 hours

2021 version 1

QUESTION BOOK

Structure of book

<i>Section</i>	<i>Number of questions</i>	<i>Number of questions to be answered</i>	<i>Number of marks</i>
A	20	20	20
B	10	10	50
C	3	3	30
			Total 100

SECTION A**Instructions for Section A**

Answer **all** questions.

Choose the response that is **correct** for the question.

A correct answer scores 1, an incorrect answer scores 0.

Marks will **not** be deducted for incorrect answers.

No marks will be given if more than one answer is completed for any question.

Unless otherwise indicated, the diagrams in this book are **not** drawn to scale.

Question 1

$\frac{4}{3} \div 2\frac{1}{3}$ simplifies to

- A. $\frac{4}{7}$
- B. $\frac{7}{4}$
- C. $\frac{28}{3}$
- D. $\frac{3}{28}$
- E. 4

Question 2

A bottle of shampoo, normally priced at \$8.60, is discounted by 20%.
The discounted price of the bottle of shampoo is

- A. \$1.72
- B. \$4.30
- C. \$6.40
- D. \$6.88
- E. \$8.43

The following information relates to Questions 19 – 20.

A group of 90 Year 9 students were asked about what they wished to do when they left school. The following two-way table summarises the Year 9 students' responses.

	Tertiary Education	Work
Girls	30	18
Boys	25	17

A Year 9 student from this group is randomly selected.

Question 19

The probability that this student is a girl or wanting to work is

- A. $\frac{13}{18}$
- B. $\frac{7}{18}$
- C. $\frac{8}{15}$
- D. $\frac{3}{8}$
- E. $\frac{1}{5}$

Question 20

Which one of the following statements is **true**?

- A. Randomly selecting a Year 9 student from the group is a two-step experiment
- B. The most likely outcome is that the randomly selected student is a boy who wants to work
- C. The total number of Year 9 students that were surveyed is 100
- D. The complementary event of selecting a girl is selecting a student who wishes to work
- E. It is impossible to select a Year 9 student in this group who does not wish to pursue tertiary study nor work

SECTION B**Instructions for Section B**

Answer **all** questions.

In all questions where a numerical answer is required, an **exact** value must be given unless otherwise specified.

In questions where more than one mark is available, appropriate working **must** be shown.

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Question 1 (5 marks)

a. Evaluate $8 - 17$.

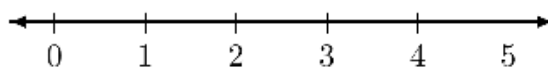
1 mark

b. Write the numbers 23.5% , $\frac{11}{43}$ and 0.25 in ascending order.

2 marks

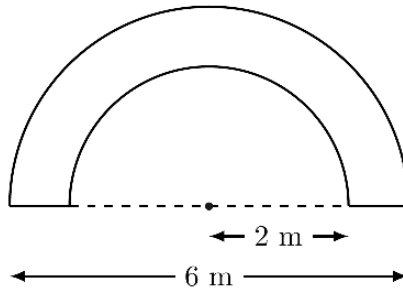
c. The number 2 is increased by 100%.
Indicate the resulting number on the number line below.

2 marks



Question 8 (5 marks)

Consider the shape shown below.



- a. The area of the shape is $k\pi \text{ m}^2$, where k is a positive constant.
Find the value of k .

3 marks

- b. Find the perimeter of the shape.
Round your answer to one decimal place.

2 marks

Question 10 (5 marks)

The table below shows the cost of electricity each day for a gym last week.

Day	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Cost (\$)	106	96	86	84	80	79	71

a. Find the median daily cost of electricity last week.

1 mark

b. Find the mean daily cost of electricity last week.

2 marks

c. Briefly describe how the cost of electricity varied each day last week.

1 mark

d. State the name of a statistical plot appropriate for displaying the data above.

1 mark

END OF SECTION B

SECTION C**Instructions for Section C**

Answer **all** questions.

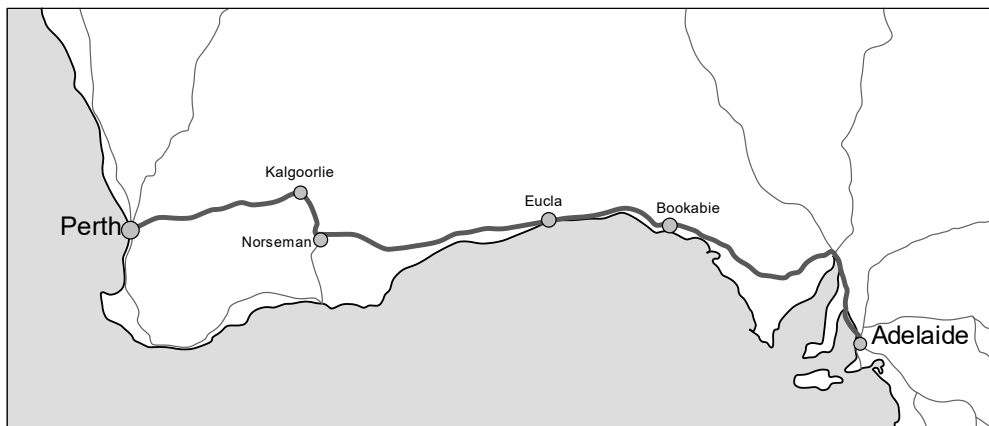
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In questions where more than one mark is available, appropriate working **must** be shown.

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Question 1 (10 marks)

The distance between Adelaide and Perth is approximately 2,696 km.



- a.** Find the distance between Adelaide and Perth in metres.
Show your working.

2 marks

- b.** Express your answer to part **a** in scientific notation.

1 mark

- c.** A traveller is driving from Adelaide to Perth.
Kalgoorlie is 80.16% of the way to Perth from Adelaide.
What is the distance between Kalgoorlie and Perth?
Round your answer to the nearest kilometre.

2 marks

Question 3 (10 marks)

Two pharmacies (Downtown and Midtown) both sell a particular brand of sunscreen ahead of summer. The number of bottles of sunscreen sold at these two pharmacies over a 13-day period are compared in the following back-to-back stem-and-leaf plot.

Downtown						Midtown				
				9	0	2	5	6		
		8	6	5	1	1	1	3	5	7 8
	8	7	4	4	2	3	5	5	9	
9	8	5	4	2	3					

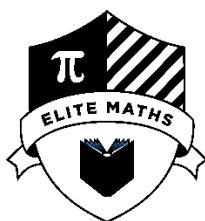
0|2 means 2 bottles

- a.** What type of variable is the number of bottles sold? 1 mark

- b.** Show that the range of bottles sold for Midtown pharmacy is 27. 1 mark

- c.** Which shop has a greater spread of the number of bottles sold? 2 marks
Support your answer with a relevant measure.

- d.** On what percentage of days did Downtown pharmacy sell more than 18 bottles? 2 marks
Round your answer to one decimal place.



2021 YEAR 9 (5.2) MATHEMATICS

Written examination

Reading time: 15 minutes

Writing time: 3 hours

2021 version 1

QUESTION BOOK

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<i>Section</i>	<i>Number of questions</i>	<i>Number of questions to be answered</i>	<i>Number of marks</i>
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			Total 100

Question 9

The equation of a circle centred at the origin with a radius of 3 is

- A. $x^2 + y^2 = 6$
- B. $x^2 + y^2 = 9$
- C. $x^2 + y^2 = 3$
- D. $x^2 + y^2 = 1$
- E. $x^2 + y^2 = 2$

Question 10

Some working for solving the equation $\frac{2m}{3} - \frac{m+1}{2} = -1$ is shown below.

Line 1: $6 \times \left(\frac{2m}{3} - \frac{m+1}{2} \right) = 6 \times (-1)$

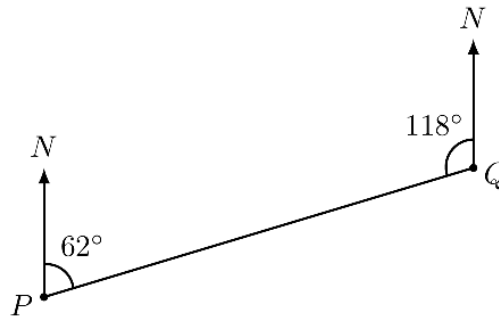
Line 2: $4m - 3(m+1) = -6$

Line 3: $4m - 3m + 3 = -6$

Line 4: $m = -9$

Which one of the following statements is **true**?

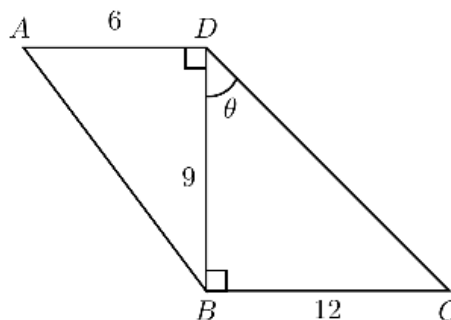
- A. The first error was made in **Line 1**
- B. The first error was made in **Line 2**
- C. The first error was made in **Line 3**
- D. The first error was made in **Line 4**
- E. There were no errors and $m = -9$ is the correct solution

Question 11

Which one of the following statements is **true** about the diagram above?

- A. The bearing of P from Q is 242°
- B. The bearing of P from Q is 118°
- C. The bearing of P from Q is 062°
- D. The bearing of Q from P is 118°
- E. The bearing of Q from P is 242°

The following information relates to Questions 12 – 13.

**Question 12**

The value of θ is closest to

- A. 33.7°
- B. 36.9°
- C. 41.4°
- D. 41.8°
- E. 53.1°

Question 13

The perimeter of the quadrilateral $ABCD$ is closest to

- A. 16.8
- B. 43.8
- C. 81.0
- D. 162
- E. 360

The following information relates to Questions 19 – 20.

Two letters are randomly selected from the word SIT.

Question 19

Which of the following tables correctly represents the outcomes if the experiment is conducted **without replacement**?

A.

		1st		
2nd		S	I	T
	S	×	(I,S)	(T,S)
	I	(S,I)	×	(T,I)
	T	(S,T)	(I,T)	×

B.

		1st		
2nd		S	I	T
	S	(S,S)	(I,S)	(T,S)
	I	(S,I)	(I,I)	(T,I)
	T	(S,T)	(I,T)	(T,T)

C.

		1st		
2nd		S	I	T
	S	×	(S,I)	(S,T)
	I	(I,S)	×	(I,T)
	T	(T,S)	(T,I)	×

D.

		1st		
2nd		S	I	T
	S	(S,S)	(S,I)	(S,T)
	I	(I,S)	(I,I)	(I,T)
	T	(T,S)	(T,I)	(T,T)

E.

		1st		
2nd		S	I	T
	S	(S,S)	×	×
	I	×	(I,I)	×
	T	×	×	(T,T)

END OF SECTION A

SECTION B**Instructions for Section B**

Answer **all** questions.

In all questions where a numerical answer is required, an **exact** value must be given unless otherwise specified.

In questions where more than one mark is available, appropriate working **must** be shown.

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Question 1 (5 marks)

a. Use your calculator to evaluate $\sqrt{1.69}$.

1 mark

b. Find the result of increasing \$620 by 15%.

2 marks

c. Write $\frac{53}{66}$ as a recurring decimal.

2 marks

Question 3 (5 marks)

There are 2^{15} users of a mobile application today.

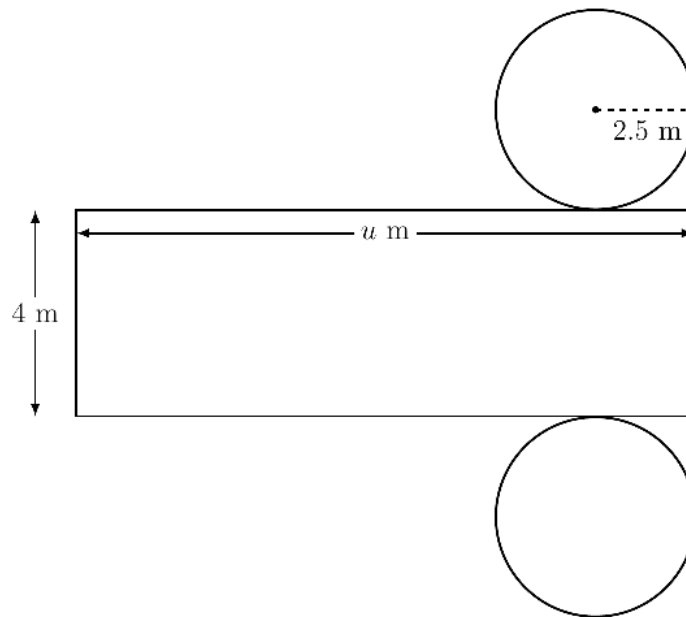
Due to popularity, the number of users doubles every three days.

- a. Write down the number of users of the mobile application 6 days from today in the form 2^n , where n is a positive integer. 2 marks

- b. After how many days will there be 2^{22} users? 3 marks

Question 7 (5 marks)

A net of a cylinder is shown below.



- a. Write down an expression for u in terms of π .

1 mark

- b. Find the surface area of the cylinder.
Round your answer to one decimal place.

2 marks

- c. Two of these cylinders are stacked on top of one another.
Find the combined volume of the cylinders.
Round your answer to one decimal place.

2 marks

SECTION C**Instructions for Section C**

Answer **all** questions.

In all questions where a numerical answer is required, an **exact** value must be given unless otherwise specified.

In questions where more than one mark is available, appropriate working **must** be shown.

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Question 1 (10 marks)

Elly drove from a farm to the city at an average speed of $16\frac{2}{3}$ metres per second.

She drove back to the farm at an average speed of 80 kilometres per hour.

Elly's trip took 3.5 hours in total.

- a.** Show that $16\frac{2}{3}$ metres per second is equal to 60 kilometres per hour. 2 marks

- b.** Round $16\frac{2}{3}$ to two decimal places. 1 mark

The distance between the farm and the city is d kilometres.

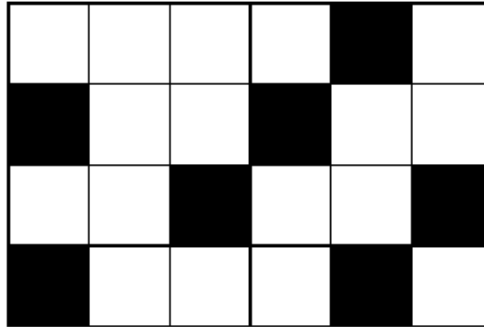
- c.** Write each of the following expressions in terms of d .

- i.** The time, in hours, taken by Elly to travel from the farm to the city. 1 mark

- ii.** The time, in hours, taken by Elly to travel from the city back to the farm. 1 mark

Question 3 (10 marks)

A grid is made up of black squares and white squares.



A square from this grid is randomly selected and its colour is noted.

This experiment is repeated 1,000 times.

The following table is incomplete as it only shows the number of black squares obtained.

Outcome	Frequency
Black	303
White	
Total	

a. Complete the table by writing down the appropriate numbers.

1 mark

b. Find the experimental probability of obtaining a black square.

2 marks

Write your answer as a decimal.

c. Is the theoretical probability of obtaining a black square close to the experimental probability?

2 marks

Justify your answer.
